

Applied Physics

Credit Hours: Three 3(2-1)

Course Code: PHYS-103

Course Outlines:

Electrostatics: Electric Charge, Conductors and Insulators, Charge in isolated conductors, Electric Fields due to a Point Charge, Electric Flux, Gauss' Law and its Applications, Electric Potential, Equipotential Surfaces, Potential due to a Point Charge and a Group of Point Charges. Relation between Electric Field and Electric Potential, Field as the gradient or derivative of potential.

Current and Electricity: Electric Current and Current Density, Resistance and Resistivity, conductors and conductivity σ , Ohm's Law, Power in Electric Circuits, Magnetic force on charge particle and current carrying wire, magnetic field lines, Biot-savart law, Two parallel conductors. Amperes' s Law, Solenoid, Toroids, Faraday's experiments, Faraday's Law of Induction, Lenz's law, Motional emf, Induced electric field, The basic equation of electromagnetism, Induced Magnetic field, The displacement current.

Optics: Refraction of light waves, Total internal reflection, Two source interference, Double Slit interference, related problems, Interference from thin films, Diffraction and the wave theory, related problems, Single-Slit Diffraction, related problems, Polarization of electromagnetic waves, Polarizing sheets, related problems

Lab:

To study the resistor coding and combination of resistors by using resistors of different resistances, To find the resistance of a wire by using Wheat stone bridge, conversion of a galvanometer into a voltmeter and ammeter, Calibration of ammeter and voltmeter by using potentiometer.

Experiments demonstrating optical phenomena such as interference, diffraction, linear motion, reflection, refraction, dispersion. Michelson interferometry, measurement of refractive index using interferometry, measurement of the speed of light, diffraction gratings and multiple-slit interference.